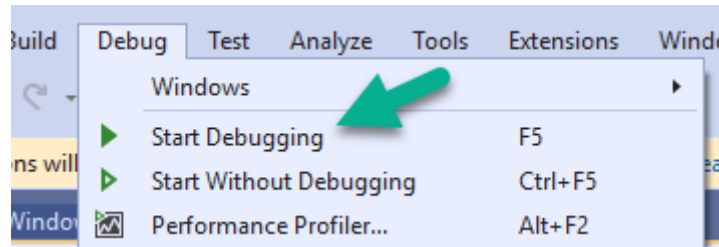


Debugging guide v1

The process of solving code bugs is called Debugging. This basically means investigating the program's execution and state

Debugging in Visual Studio occurs automatically when you run from Visual Studio with F5 or select **Debug | Start Debugging**.



Break points:

You set breakpoints wherever you want to pause debugger execution. For example, you may want to see the state of code variables or look at the call stack at a certain breakpoint.

To set a breakpoint in source code, click in the far left margin next to a line of code. You can also select the line and press **F9**, select **Debug > Toggle Breakpoint**, or right-click and select **Breakpoint > Insert breakpoint**. The breakpoint appears as a red dot in the left margin.

To disable a breakpoint without deleting it, hover over or right-click it, and select **Disable breakpoint**. Disabled breakpoints appear as empty dots in the left margin or the **Breakpoints** window. To re-enable a breakpoint, hover over or right-click it, and select **Enable breakpoint**.

Steps to debug a simple code (basic level)

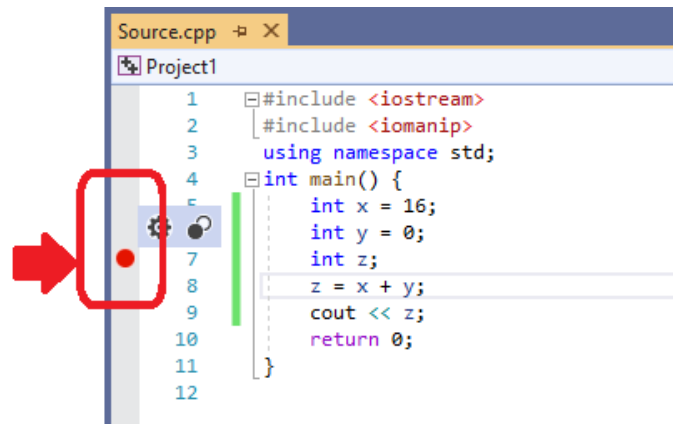
Step 1

Create a main cpp file (by creating new project) and paste the following code.

```
#include <iostream>
using namespace std;
int main() {
    int x = 16;
    int y = 0;
    int z;
    z = x + y;
    cout << z;
    return 0;
}
```

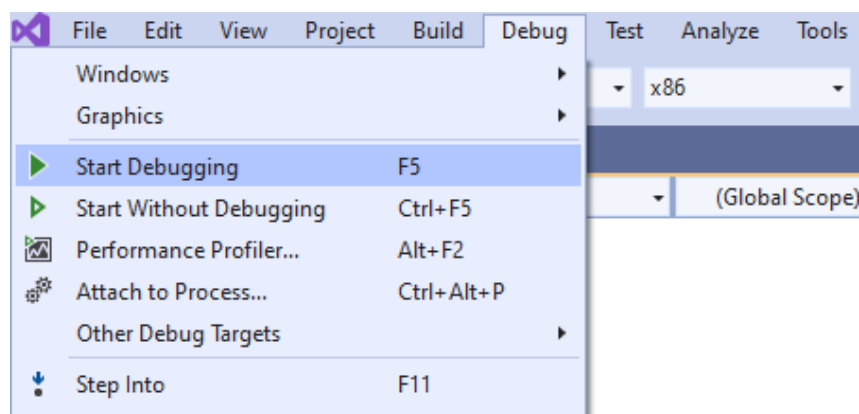
Step 2

You need to toggle a break point (according to the instructions mentioned above at the start). As you can see the following screenshot, you can click on the left most side of the line to toggle a break point directly. Once toggled, red filled circle appears there.



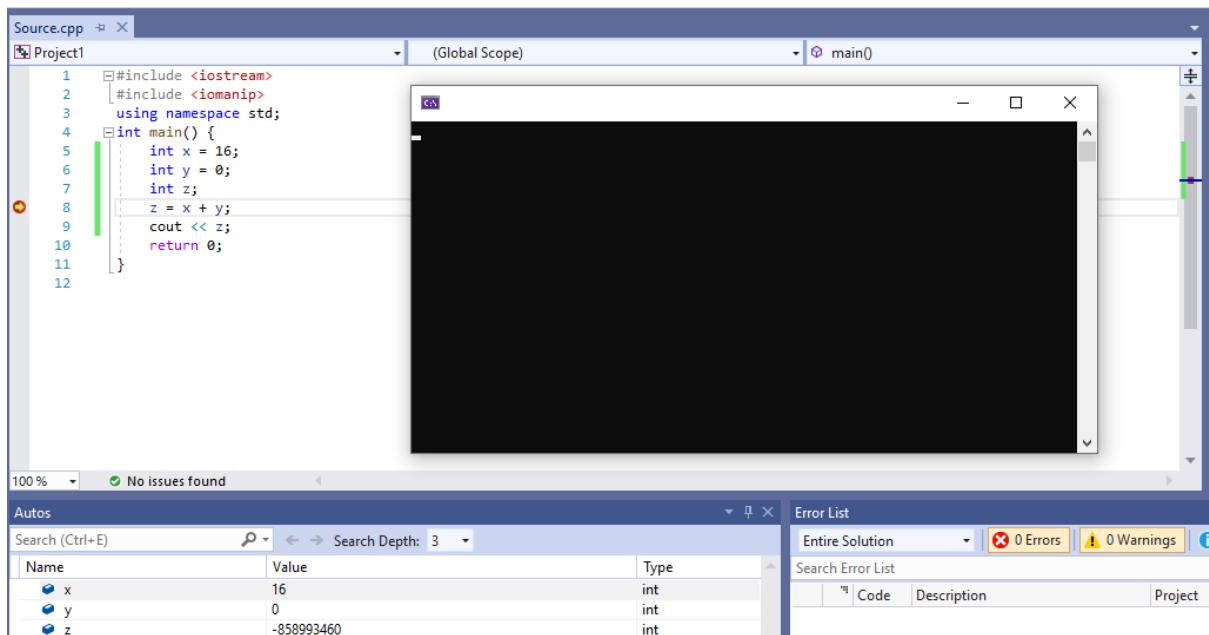
Step 3

Next move to **Debug-> Start Debugging**



As you have toggled a break point at line# 7 therefore execution of the program stops at this line.

The explorer and console at this point is shown below

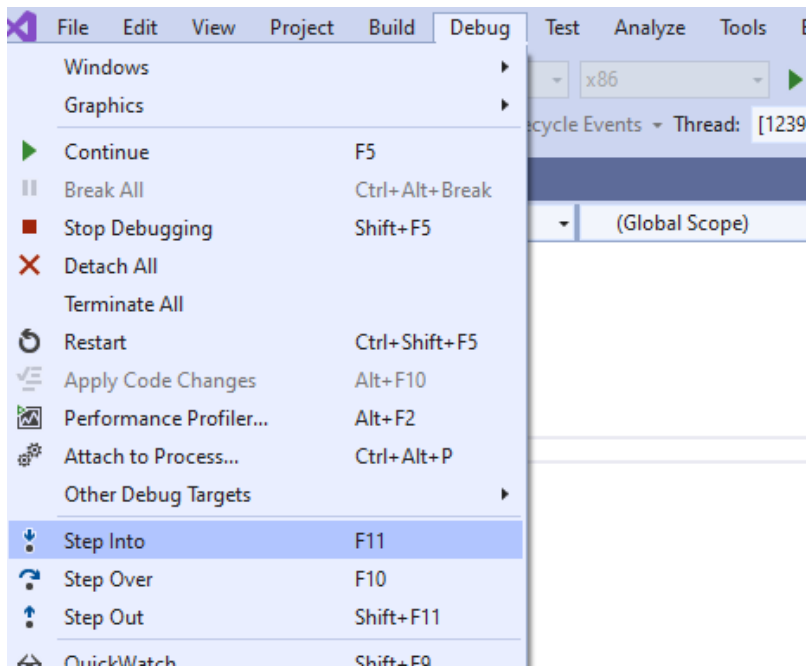


You can notice a small yellow arrow sign at that break point line. At this point, the program is executed till line# 7. Till line# 7, the value of variables are shown as below. Notice that variable z is only declared at line#7 therefore it has some garbage value. Nothing has been shown in console window yet because your code (till line#7) does not has any cout statement.

Search (Ctrl+E)			Search Depth: 3
Name	Value	Type	
x	16	int	
y	0	int	
z	-858993460	int	

Step 4:

Proceed with the code execution. For that, go to **Debug->Step into** or press **F11**.



Once you do that following changes are made to the variables. You'll notice that arrow now moves to line#8. Again press F11 to execute this current line. The value of variables will be updated as shown below

```

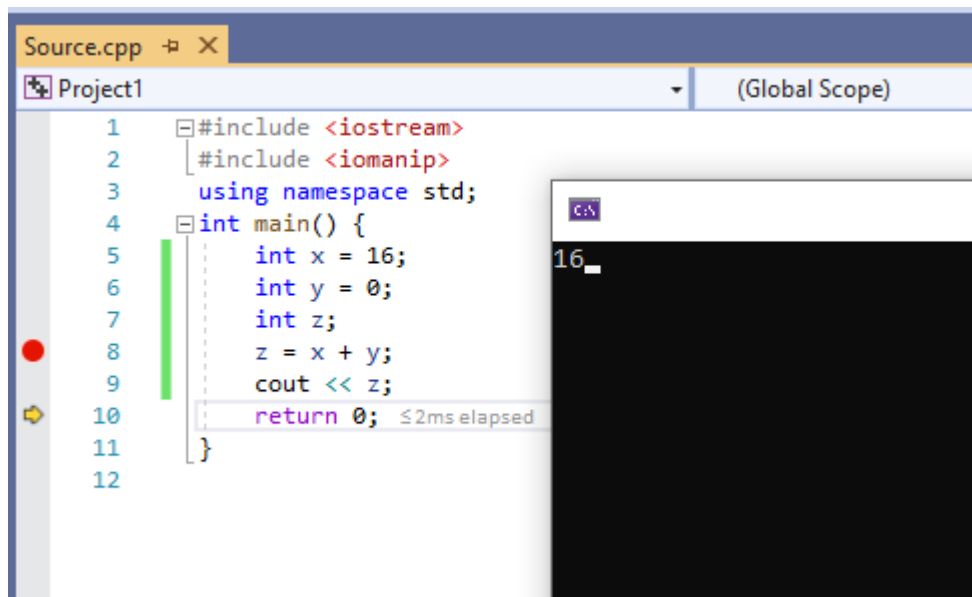
1  #include <iostream>
2  #include <iomanip>
3  using namespace std;
4  int main() {
5      int x = 16;
6      int y = 0;
7      int z;
8      z = x + y;
9      cout << z;
10     return 0;
11 }
12

```

Name	Value	Type
x	16	int
y	0	int
z	16	int

Since line#8 updates the value of z, therefore z is updated with the value 16. Console screen is still empty due to same reasons as discussed.

Proceed forward and again press F11. Now, line# executes a cout statement therefore console will reflect this change.



The screenshot shows a C++ IDE with a source file named 'Source.cpp' and a project named 'Project1'. The code in the source file is as follows:

```
1 #include <iostream>
2 #include <iomanip>
3 using namespace std;
4 int main() {
5     int x = 16;
6     int y = 0;
7     int z;
8     z = x + y;
9     cout << z;
10    return 0;
11 }
12
```

The output window on the right shows the result of the program execution, which is '16'. A status bar at the bottom of the source file indicates that the program executed successfully in less than 2ms.

Note: If you choose **Continue** option from the top toolbar then the program does not executes line wise one by one. Instead, it executes in one go (as it does when we run without debugging).